

Database Normalization

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Normalization is a systematic technique used in relational database design to organize data into well-structured tables.

The main goal are to:

- Reduce data redundancy.
- Prevent insertion anomalies.
- Prevent update anomalies.
- Prevent deletion anomalies.
- Improve data integrity.
- To make dependencies between Data clear.
- Produce a table that are easier to maintain.

The normal forms are usually studied in this order:

```
UNF
|
v
1NF
|
v
2NF
|
v
3NF
|
v
BCNF
|
v
4NF
|
v
5NF
```

Each normal form imposes additional condition on the database design.

Why Normalization is Required?

Consider this table:

	Student_id	Name	Course_id	Course_name	Instructor
▶	101	Alice Johnson	C101	Database Management	Dr. Smith
	101	Alice Johnson	C102	Data Structures	Prof. Brown
	101	Alice Johnson	C103	Computer Networks	Dr. Wilson
	102	Bob Williams	C101	Database Management	Dr. Smith
	102	Bob Williams	C104	Operating Systems	Prof. Taylor
	103	Charlie Davis	C105	Software Engineering	Dr. Anderson
	103	Charlie Davis	C106	Web Development	Prof. Thomas
	104	Diana Miller	C102	Data Structures	Prof. Brown
	105	Ethan Clark	C107	Machine Learning	Dr. Garcia
	105	Ethan Clark	C108	Cloud Computing	Prof. Martin

There is redundancy.

For example:

Student_id = 101

Name = Alice Johnson

Is repeated.

Similarly:

C101 -> Database Management->Dr.Smith
is repeated for every student taking C101.

C102 ->Data Structures -> Prof.Brown
is repeated for every student taking C102.

This can produce several problems.